```
import cv2
import numpy as np
from google.colab import files
import matplotlib.pyplot as plt

# Upload image
uploaded = files.upload()

# Get uploaded file name
image_path = list(uploaded.keys())[0]

# Read image
img = cv2.imread(image_path)

# Check if image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Get image dimensions
    rows, cols = img.shape[:2]

    # Translation matrix (move image 100 pixels right and 50 pixels down)
    M = np.float32([[1, 0, 100],
                    [0, 1, 50]])

    # Apply affine transformation
    affine_img = cv2.warpAffine(img, M, (cols, rows))

    # Save output image
    cv2.imwrite("Affine_Transformed.jpg", affine_img)

    # Convert BGR to RGB for displaying
    original_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    transformed_rgb = cv2.cvtColor(affine_img, cv2.COLOR_BGR2RGB)

    # Display images
    plt.figure(figsize=(10,5))

    plt.subplot(1,2,1)
    plt.imshow(original_rgb)
    plt.title("Original Image")
    plt.axis("off")

    plt.subplot(1,2,2)
    plt.imshow(transformed_rgb)
    plt.title("Affine Transformed Image")
    plt.axis("off")

    plt.show()
```

```
# Download the transformed image
files.download("Affine_Transformed.jpg")
```

Choose Files sampled_image.jpg

sampled_image.jpg(image/jpeg) - 9699 bytes, last modified: 7/22/2026 - 100% done

```
Saving sampled_image.jpg to sampled_image (2).jpg
```

Original Image

Affine Transformed Image